

Sliding Window protocols Summary With Questions

GeeksforGeeks

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Sliding Window Protocol is a flow control mechanism used in data link and transport layers to manage the reliable and efficient transmission of frames or packets between sender and receiver.

Key Concepts in Sliding Window Protocol :

Window Size: Refers to the number of frames that can be sent before requiring an acknowledgment.

Sender Window: Controls how many frames the sender can transmit before pausing for an ACK.

Receiver Window: Indicates how many frames the receiver is ready to accept.

Acknowledgment (ACK): Receiver confirms the successful reception of frames.

Types of Sliding Window Protocols:

Stop-and-Wait: Sends one frame at a time, waits for ACK.

Go-Back-N (GBN): Sender can send several frames (window size N) without waiting, but on error, retransmits from the erroneous frame onward.

Selective Repeat (SR): Only the erroneous or lost frames are resent, not the whole sequence.

Useful time : $T_t(\text{data})$

Total Time : $T_t(\text{data}) + 2 * T_p + T_q + T_{pro} + T_t(\text{ack})$

Efficiency = $T_t(\text{data}) / T_t(\text{data}) + 2 * T_p + T_q + T_{pro} + T_t(\text{ack})$

$T_t(\text{data})$: Transmission delay for Data packet

T_p : Propagation delay for Data packet

T_q : Queuing delay

T_{pro} : Processing delay

$T_t(\text{ack})$: Transmission delay for acknowledgment

Stop And Wait

[Stop and Wait ARQ](#) is a Sliding Window Protocol method used for the reliable delivery of data frames.

The stop-and-wait ARQ is used for noisy channels or links to handle flow and error control between

sender and receiver. The Stop and Wait ARQ protocol sends a data frame and then waits for an acknowledgment (ACK) from the receiver.

Sender window size (W_s) = 1

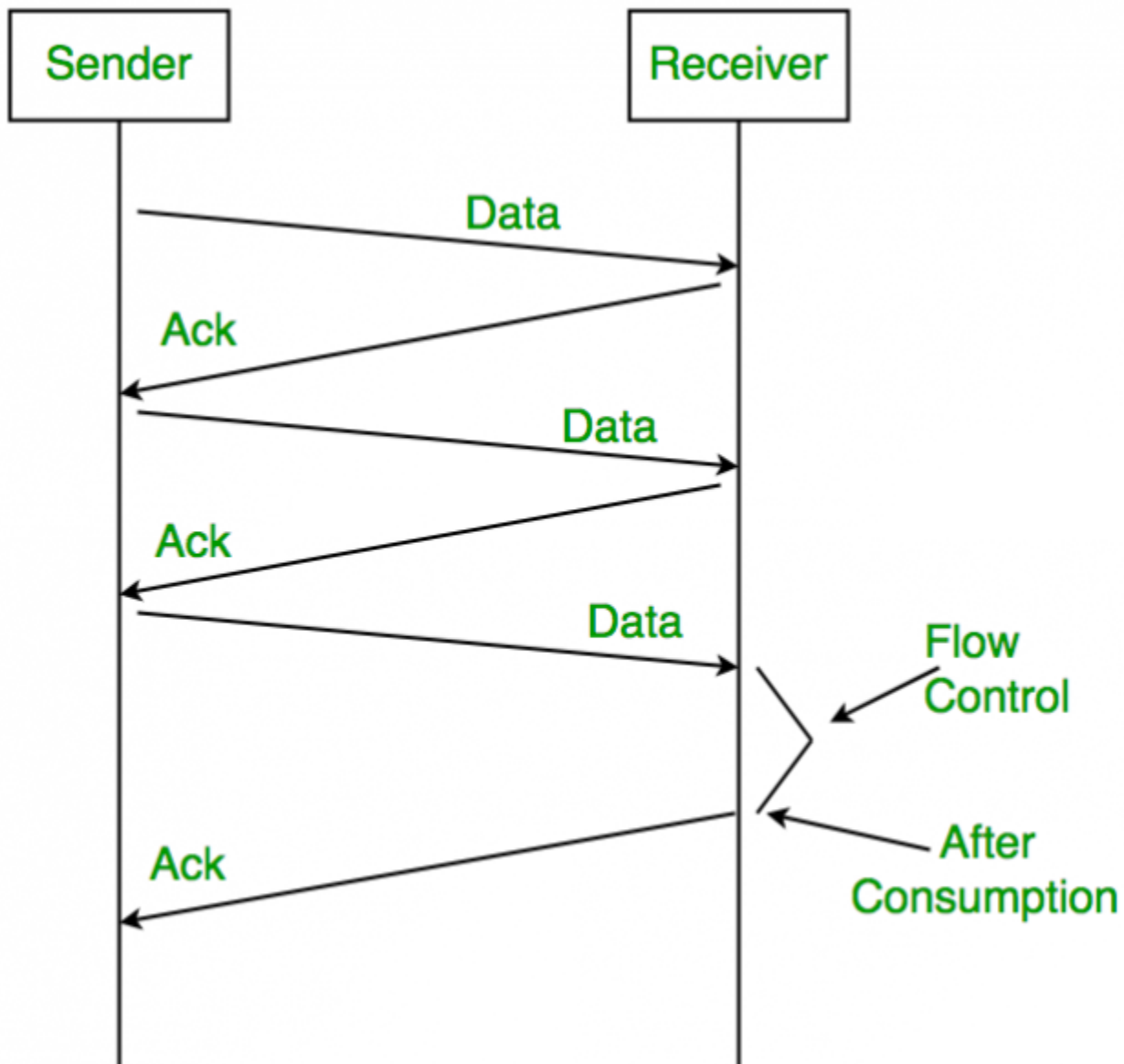
Receiver window size (W_r) = 1

Sequence Number $\geq 1 + 1$

Uses independent acknowledgement

Discards out of order packets

Efficiency= (Useful time)/(Total Time)



STOP AND WAIT WORKING

Go Back N

In [Go-Back-N](#), the sender can send multiple data packets without waiting for an acknowledgement for each one. However, it can only send a certain number of packets (this is called the window size). If one packet is lost or not acknowledged, the sender must go back and resend that packet and all the packets that followed it, even if they were received correctly.

Sender window size $W_s = N$

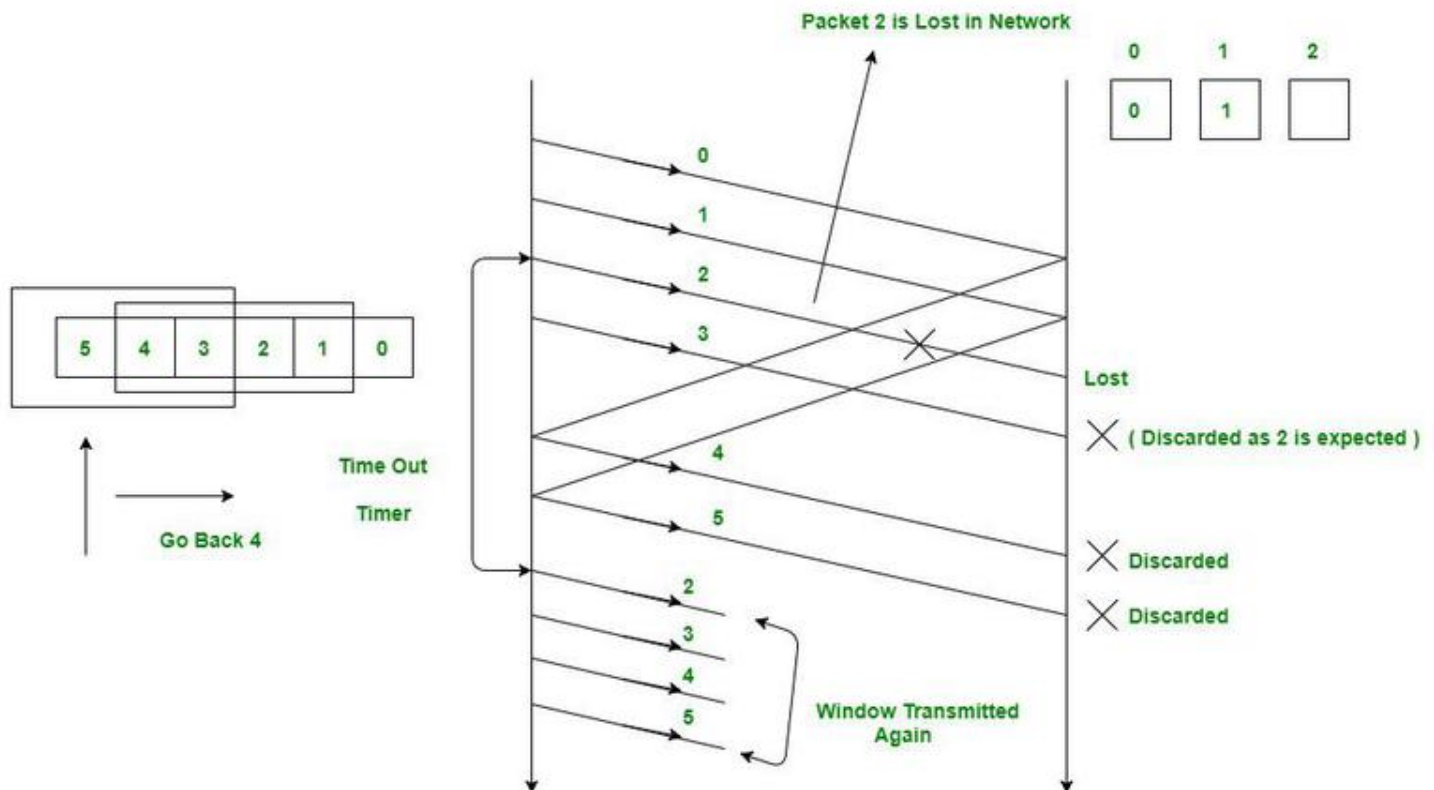
Receiver window size $W_r = 1$

Sequence number $\geq N + 1$

Can use both cumulative or independent acknowledgement depends on acknowledge timer

Discards out of order packets

Efficiency = $N \cdot (\text{Useful time}) / (\text{Total Time})$



GB-N WORKING DIAGRAM

Selective Repeat

[Selective Repeat protocol \(SRP\)](#) is mostly identical to GBN protocol, except that buffers are used and the receiver, and the sender, each maintains a window of size. SRP works better when the link is

very unreliable. Because in this case, retransmission tends to happen more frequently, selectively retransmitting frames is more efficient than retransmitting all of them. SRP also requires full-duplex link.

Sender window size $W_s = N$

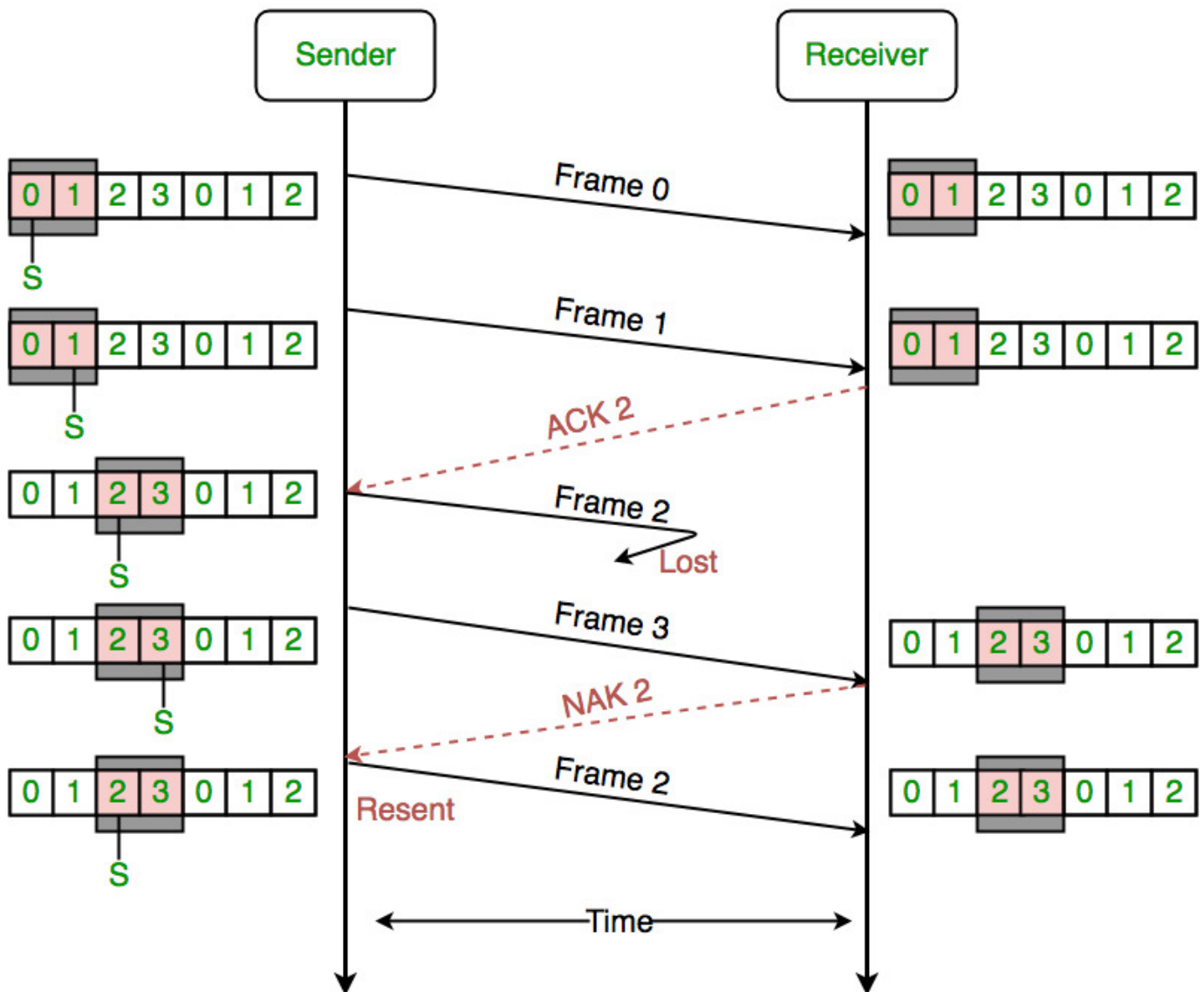
Receiver window size $W_r = N$

Sequence Number $\geq N + N$

Uses only independent acknowledgement

Can Accept out of order packets

Efficiency= $N \cdot (\text{Useful time}) / (\text{Total Time})$



SELECTIVE REPEAT WORKING

Practice Questions

Example-1. In Stop and wait protocol every 4th packet is lost and we need to send total 10 packets so how many transmission it took to send all the packets?

Explanation

1 2 3 4 5 6 7 8 9 10 (Initially)

^

1 2 3 4 4 5 6 7 8 9 10 (Packet no. 4 retransmitted)

^

1 2 3 4 4 5 6 7 7 8 9 10 (Packet no. 10 retransmitted)

^

1 2 3 4 4 5 6 7 7 8 9 10 10 (Result)

So, we retransmitted packet number 4, 7, 10 Total count = 13

Example-2. In S&W protocol if Error probability is p and no. of packets to send is ' n '. How many packets we have to send ?

Explanation

Total retransmissions = $n \cdot p^0 + n \cdot p^1 + n \cdot p^2 + n \cdot p^3 + n \cdot p^4 + \dots = n(1 + p + p^2 + p^3 + p^4 + \dots) = n \cdot (1 / (1 - p))$ using infinite GP sum formula

Example-3. In GBN sender Window size = 10 and $T_p = 49.5\text{ms}$ & $T_t = 1\text{ms}$. What is the Efficiency of the protocol and Throughput given Bandwidth = 1000 bps?

Explanation

Efficiency = $N / (1 + 2a)$ as there is only transmission delay and propagation delay, $N = 10$ (given), $a = T_p / T_t = 49.5$

Efficiency = $10 / (1 + 2 * 49.5)$

= $10 / 100$

= 0.1 or 10%

Throughput = Efficiency * Bandwidth = $0.1 * 1000 = 100$

Example-4. In GB3 if every 5th packet is lost & we need to send 10 packets so how many retransmissions are required ?

Explanation

1 2 3 4 5 6 7 | 8 9 10

^ \$ (packet no. 5 lost)

1 2 3 4 5 6 7 5 6 7 8 9 | 10

* ^ \$

1 2 3 4 5 6 7 5 6 7 8 9 7 8 9 10

* ^ \$

1 2 3 4 5 6 7 5 6 7 8 9 7 8 9 10 9 10 (count starts from * till ^)

(from ^ to \$ retransmission is done)

Note - From the Last packet is window size to lost packet we resend the entire window. Total no. of transmissions = 18

Example-5. In SR $W_s = 5$ and we are sending 10 packets where every 5th packet is lost find number of retransmissions?

Explanation

1 2 3 4 5 6 7 8 9 10

^

1 2 3 4 5 5 6 7 8 9 10

^

1 2 3 4 5 5 6 7 8 9 9 10

We see here there is no role of Window size in SR only the lost packet is resent. Total transmissions = 12

Example-6. If there is K bits sequence no. define require sender window size and receiver window size for S&W, GBN & SR?

Explanation

Given, K bits,

For S&W $W_s = 1$ and $W_r = 1$

For GBN, $W_s = 2^K - 1$ and $W_r = 1$

For SR, $W_s = 2^{K-1}$ and $W_r = 2^{(K-1)}$